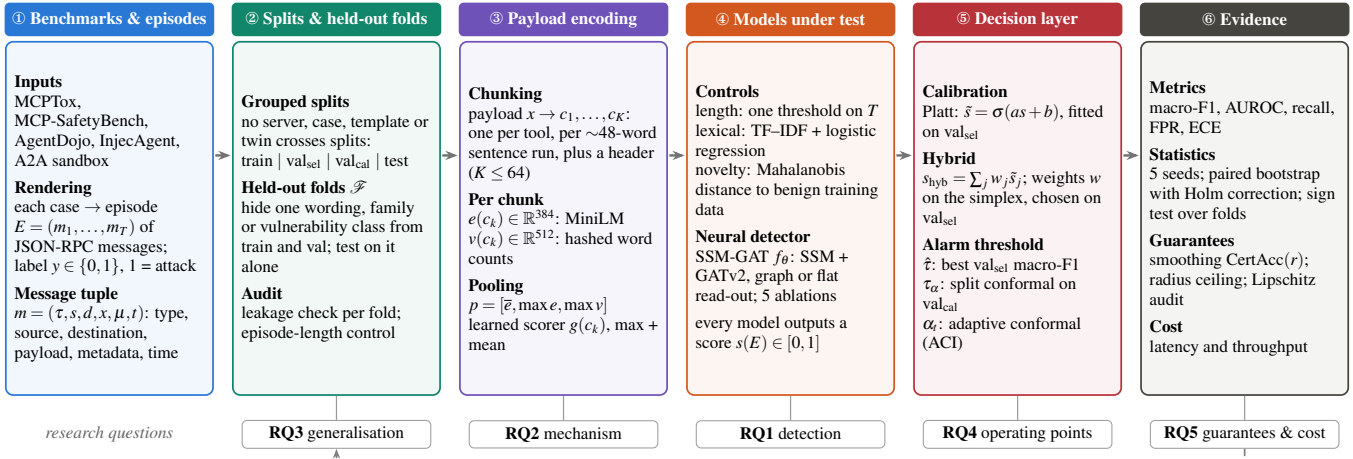

① Benchmarks & episodes
Inputs
MCPTox,
MCP-SafetyBench,
AgentDojo, InjecAgent,
A2A sandbox
Rendering
each case → episode
$E = (m_1, \ldots, m_T)$ of
JSON-RPC messages;
label $y \in \{0, 1\}$, 1 = attack
Message tuple
$m = (\tau, s, d, x, \mu, t)$: type,
source, destination,
payload, metadata, time
② Splits & held-out folds
Grouped splits
no server, case, template or
twin crosses splits:
train | val_{sel} | val_{cal} | test
Held-out folds $\mathscr{F}$
hide one wording, family
or vulnerability class from
train and val; test on it
alone
Audit
leakage check per fold;
episode-length control
③ Payload encoding
Chunking
payload $x \to c_1, \ldots, c_K$:
one per tool, per ∼48-word
sentence run, plus a header
($K \leq 64$)
Per chunk
$e(c_k) \in \mathbb{R}^{384}$: MiniLM
$v(c_k) \in \mathbb{R}^{512}$: hashed word
counts
Pooling
$p = [\bar{e}, \max e, \max v]$
learned scorer $g(c_k)$, max +
mean
④ Models under test
Controls
length: one threshold on T
lexical: TF–IDF + logistic
regression
novelty: Mahalanobis
distance to benign training
data
Neural detector
SSM-GAT f_θ: SSM +
GATv2, graph or flat
read-out; 5 ablations
every model outputs a
score $s(E) \in [0, 1]$
⑤ Decision layer
Calibration
Platt: $\tilde{s} = \sigma(as + b)$, fitted
on val_{sel}
Hybrid
$s_{\text{hyb}} = \sum_j w_j \tilde{s}_j$; weights w
on the simplex, chosen on
val_{sel}
Alarm threshold
$\hat{\tau}$: best val_{sel} macro-F1
τ_α: split conformal on
val_{cal}
α_t: adaptive conformal
(ACI)
⑥ Evidence
Metrics
macro-F1, AUROC, recall,
FPR, ECE
Statistics
5 seeds; paired bootstrap
with Holm correction; sign
test over folds
Guarantees
smoothing CertAcc(r);
radius ceiling; Lipschitz
audit
Cost
latency and throughput
research questions
RQ3 generalisation
RQ2 mechanism
RQ1 detection
RQ4 operating points
RQ5 guarantees & cost
pre-registered: every choice uses val_{sel} only; val_{cal} only sets conformal thresholds; the test split is read once